Zinc Oxide Solar Cells

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CHE - 494 Materials For Energy

Overview

- Need for Solar Cells
- Dye Synthesized Solar Cells (DSSCs)
- Zinc Oxide in DSSCs
- Construction, Working and Performance Characteristics
- Challenges
- Significance
- Proposed Research - Hypothesis
- Specific Aims
- Rationale for Proposed Study
- Scientific Impact of Study

Need for Solar Cells

- 87% of the world energy demands - Petroleum based sources
- Since the start of the 21st century - 1.8% increase in energy demand annually
- Solar Energy - 0.65% of world energy demands
- Abundant potential to compensate for and reduce non-renewable energy usage
- Carbon-neutral energy source
- Environmentally sustainable energy source

Timeline

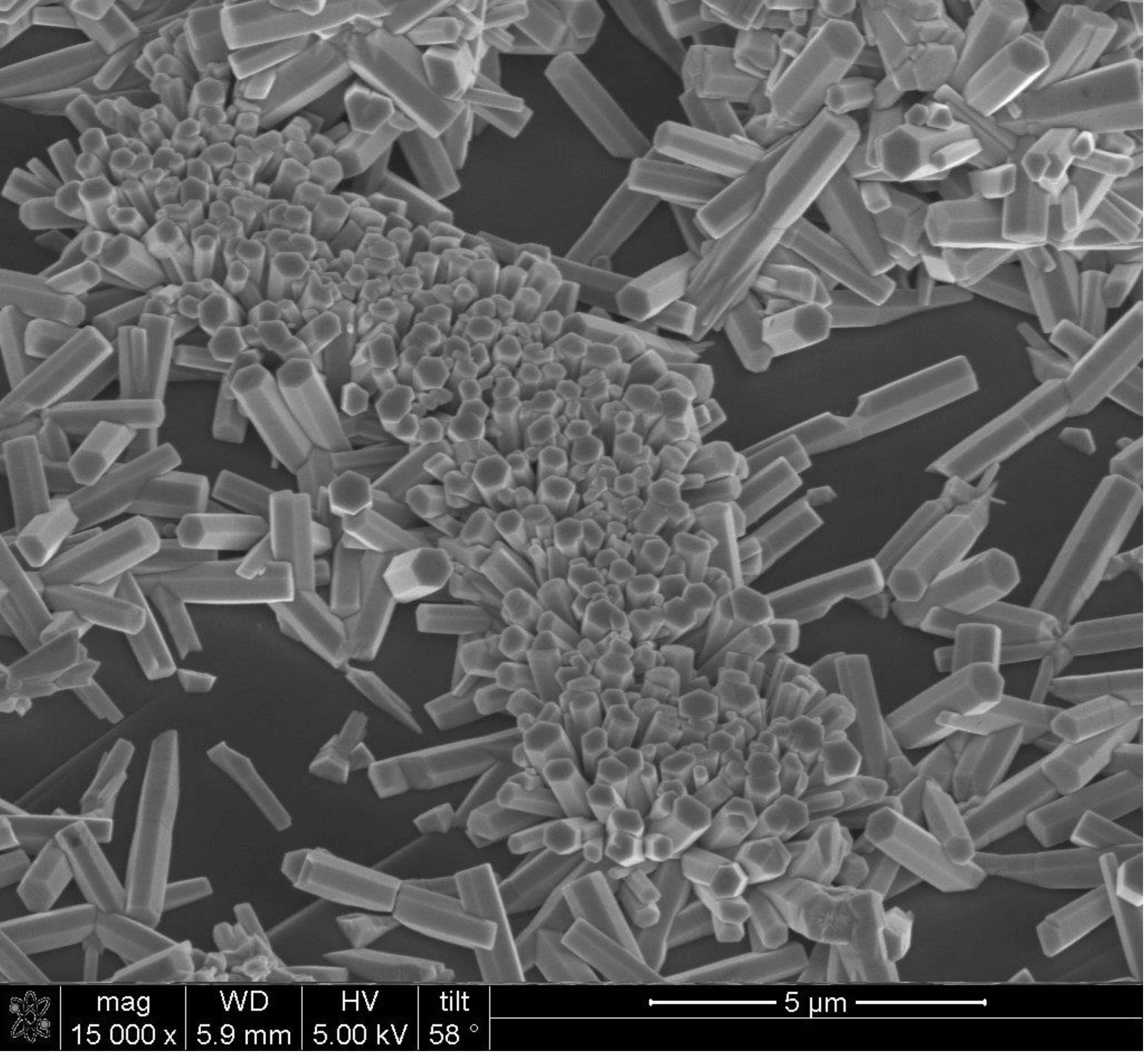
- 1839 - Discovery of Photovoltaic Effect - Alexandre-Edmond Becquerel
- 1883 - Selenium Solar Cell - Charles Fritts
- 1954 - First Practical Solar Cell - Calvin Fuller and Gerald Pearson
- 1991 - Dye Synthesized Solar Cell - Michael Graetzel and Dr Brian O’Regan

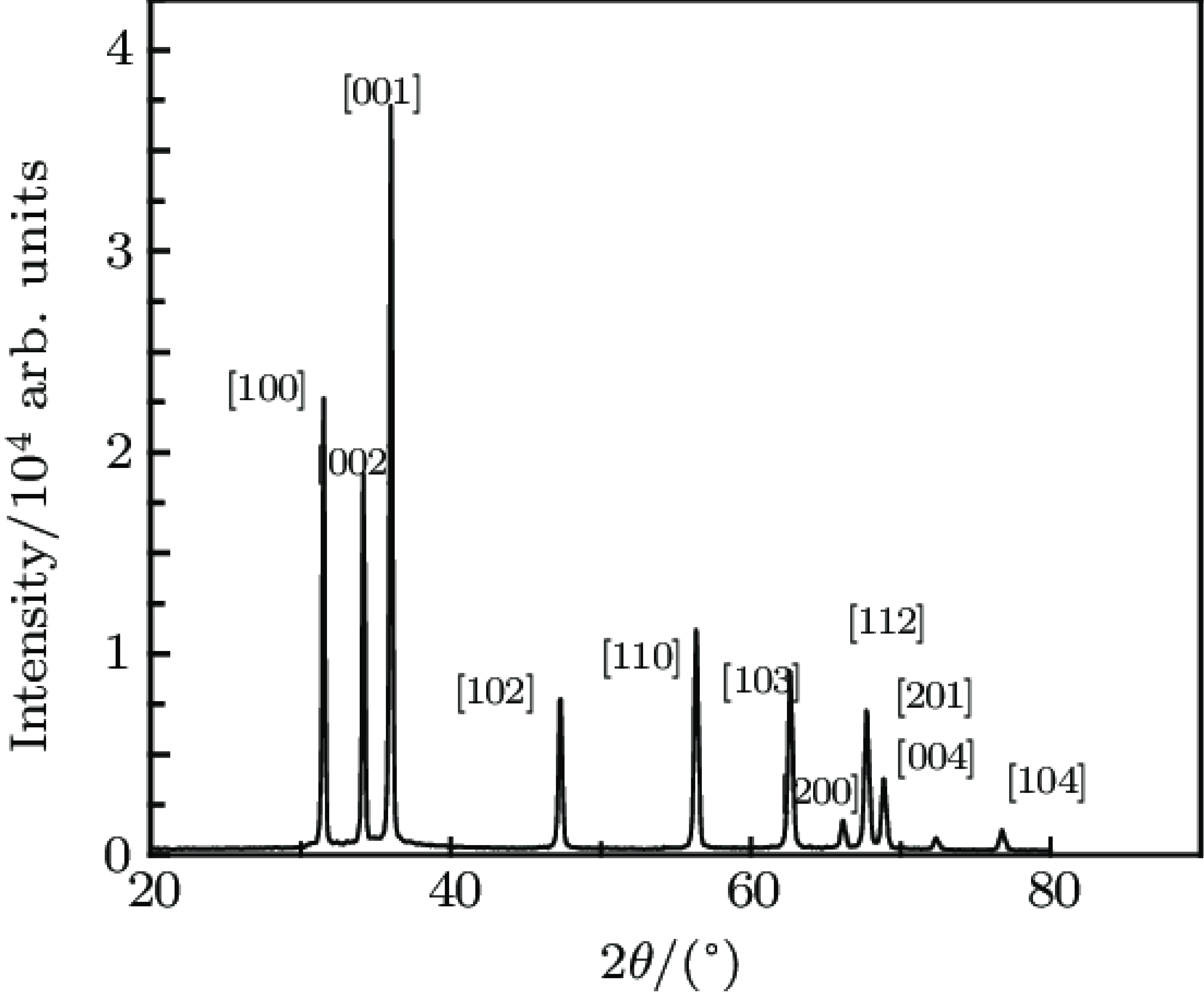
Dye Synthesized Solar Cells

- Third Generation Solar Cells
- Most cost efficient model, to date.
- Components -
  - Sensitiser (Photoactive Dye)
  - Working Electrode (deposited with a metal oxide in nano-form)
  - Redox-mediator (Chemical Electrolyte)
  - Counter electrode
- The first choice for the metal oxide was Titanium dioxide (TiO_2)
- Film Thickness - 10 microns
- Conversion Efficiency: 7.1 - 7.9%
- High Stability and Low Costs

Zinc Oxide

- Inorganic Metal Oxide occurring naturally as Zincite
- Excellent light absorbing properties in the UVA and UVB spectrum
- N-type semiconductor with a wide band gap of ~3.37 eV
- Less Toxicity than other group II-VI semiconductors
- Excellent Optical Transmittance
- Ability to be synthesised in large quantities at low costs
- Rich variety of various nano-structural forms
    - Nano-rods
    - Nano-tubes
    - Thin Films
    - Nano-wires





Scanning Electron Microscopy of Zinc Oxide Nanorods

X-Ray Diffraction Pattern of Zinc Oxide Nanorods

PP Mahamuni *et al.* Biochem Biophys Rep. 2019

Construction of Zinc Oxide Dye-Synthesised Solar Cells

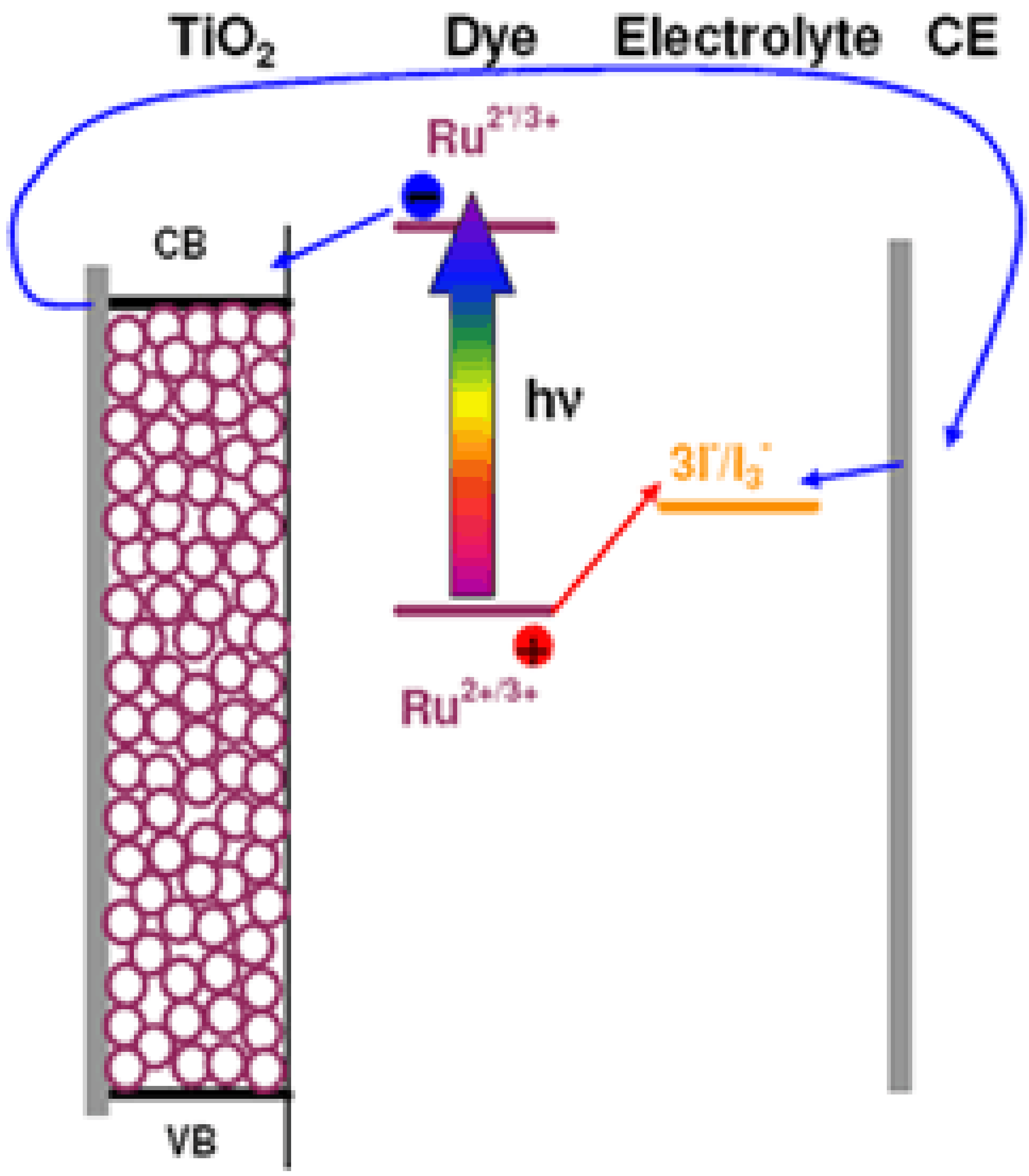
Sensitiser (Photoactive Dye)

- Acts like a molecular electron pump
- Absorbs Visible Light
- Pumps electron in conduction band of the semiconductor layer
- Accepts electron from the redox couple

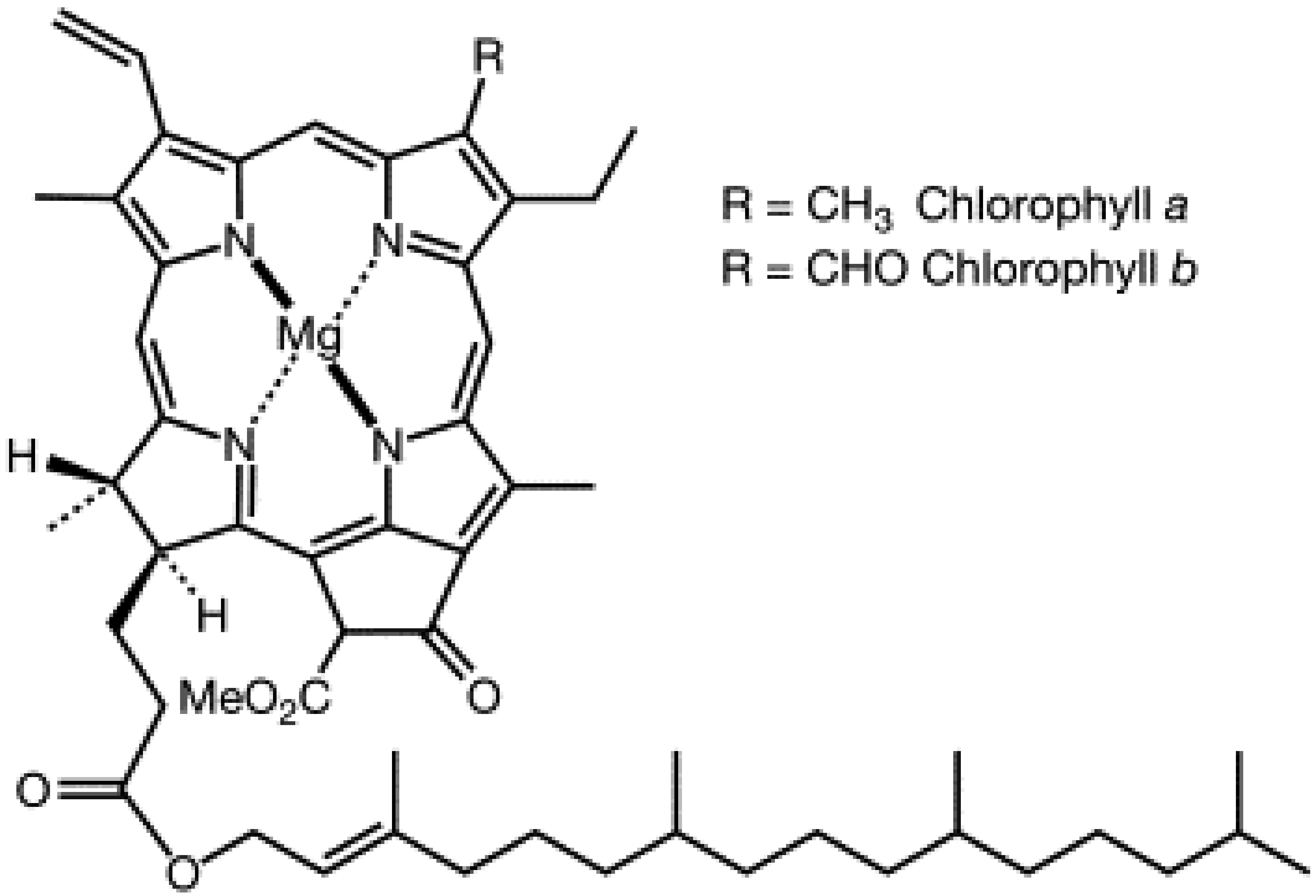
- *Strong Absorbance in the Visible Region*
- *High Stability in Ground, Oxidized and Excited States*
- *Compatible Redox Potential*
- *Good efficiency in charge injection and regeneration*

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Sensitiser (Photoactive Dye)



Ruthenium-based Dyes



Natural Dyes - Chlorophyll

Hans Desilvestro *et al* Appl. Phys. 2006

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Working Electrode

- Two sheets of transparent, conductive materials - Glass deposited with conductive substrate
- Medium for deposition of semiconductor, catalyst and act as current carriers
- Characteristics
    - High Transparency (>80%)
    - High Magnitude of Electrical Conductivity
- Reduction in Energy Loss
- Enhancement of Electron Transfer
- Material used-

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Working Electrode

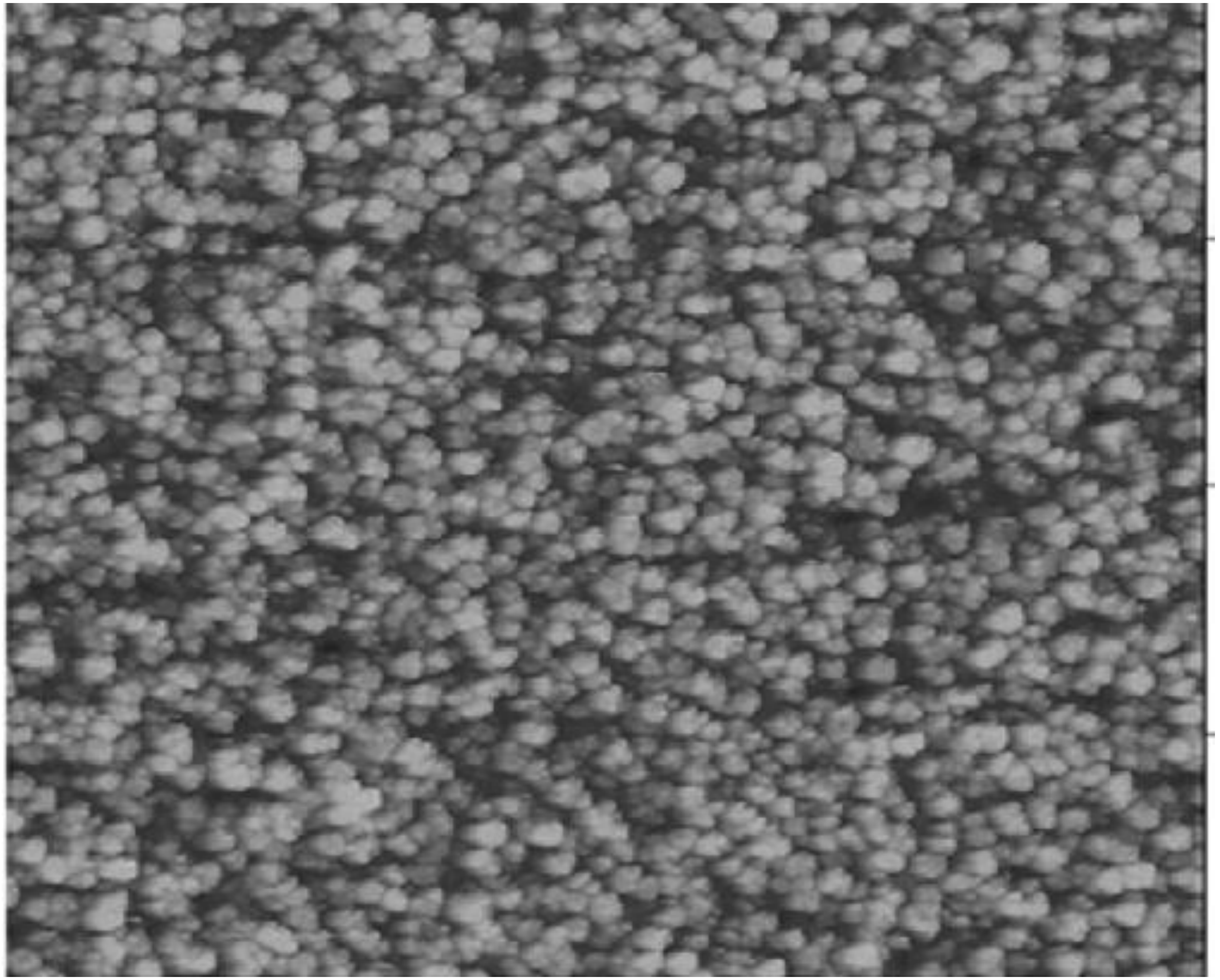
- Soda lime glass deposited with either Fluorine Doped Tin Oxide or Indium Doped Tin Oxide
  - Fluorine Doped Tin Oxide - Transmittance (75%), Sheet Resistance of 18 Ω/cm^2
  - Indium Doped Tin Oxide - Transmittance (80%), Sheet Resistance - 8 Ω/cm^2
- Thin layer of semiconductor is deposited on the working electrode
  - Zinc Nanorods
  - Thickness - 10 to 100 microns
  - Absorbance on ZnO is enhanced by immersion of electrode in the photosensitive dye
- Immersion Time -

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Working Electrode

Soda Lime Glass deposited

with Indium-doped tin oxide



SEM image of ZnO deposited Working Electrode

AC Mohan *et al.* Biosensors 2016

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Working Electrode

- Covalent bonding of dye molecules on porous surface of metal oxide
- High effective surface area of Zinc nanorods - High rate and efficiency of photon absorption
- Highest Occupied Molecular Orbital (HOMO) should be located far away from the conduction band surface of ZnO
- Lowest Unoccupied Molecular Orbital (LUMO) should be closer to ZnO

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Redox-Mediator (Chemical Electrolyte)

- Enhances reaction rate, minimises energy loss thereby increasing efficiency
- Absorption spectra of the electrolyte does not overlap with that of the dye
- Excellent, chemical, thermal and electrochemical stability
- Regeneration of oxidised dye
- Non corrosive to the glass films
- Examples - Acetonitrile (ACN) and N-methylpyrrolidine (NMP)
- ACN/valeronitrile are used owing to their high dielectric constant

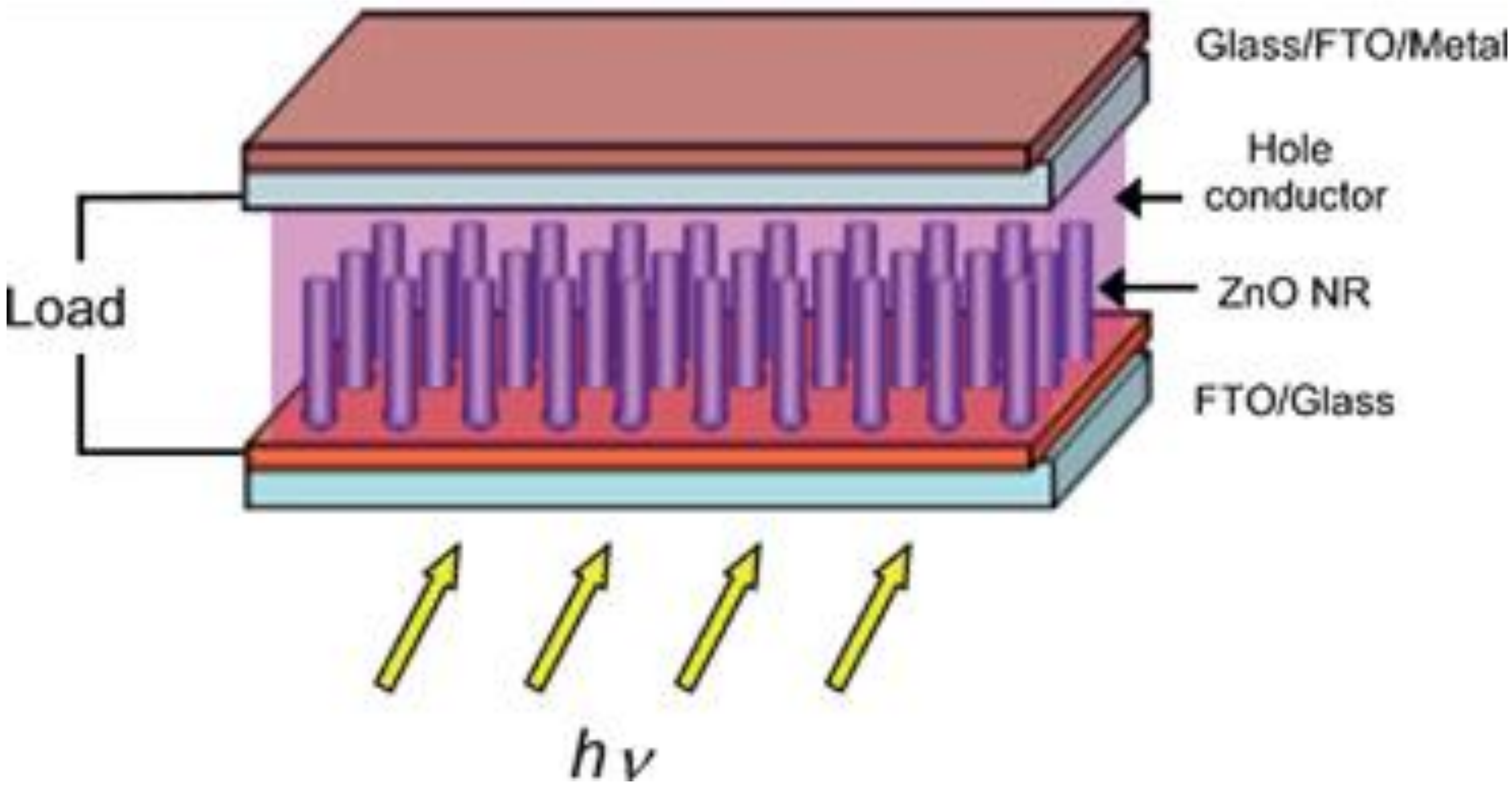
Construction of Zinc Oxide Dye-Synthesised Solar Cells

Counter Electrode

- Platinum Rod
- Connected to the working electrode, filled with electrolyte and sealed
- Completes the reaction, restoring ground state
- Facilitates continuous transfer of electrons from working electrode, to electrolyte and counter electrode
- High Efficiency
- High Costs

Construction of Zinc Oxide Dye-Synthesised Solar Cells

Schematic Diagram



- Working Electrode - Glass/Flourine doped Tin Oxide
- Photosensitive Dye - Ruthenium Dye
- Electrolyte - N-methylpyrrolidine (NMP)
- Counter Electrode - Graphite

Working of Zinc Oxide Dye-Synthesised Solar Cells

Step 1: Absorption of Photon

- Absorption of photons by the photosensitive dye
- Excitation of electrons
- Energy of photons - 1.72 eV

Step 2: Ejection of Electron

- Injection of photon in the conduction band of Zinc Oxide nanorods
- Absorption of photons from the UV region
- Oxidisation of dye

Step 3: Transport of Carrier

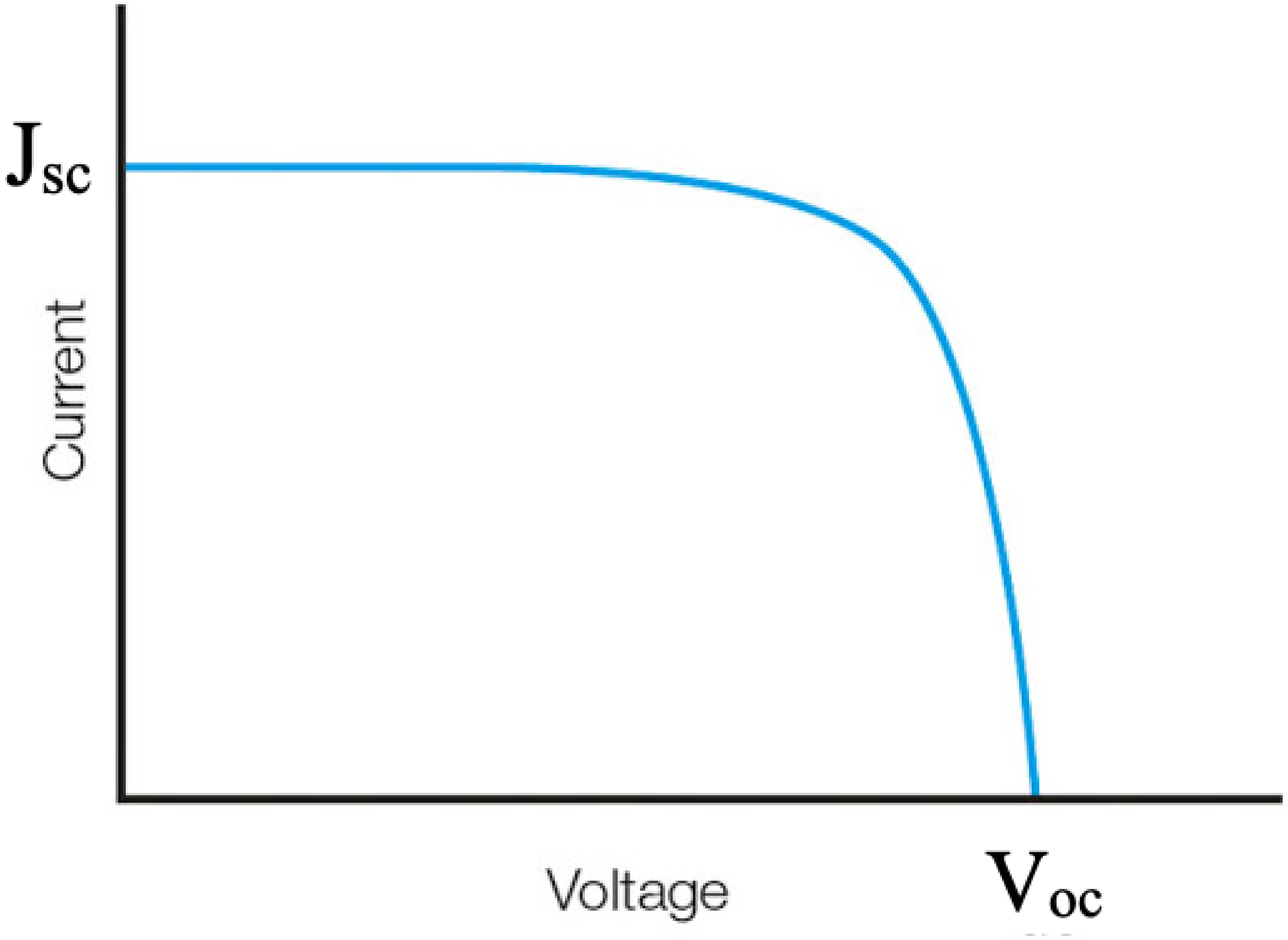
- Transport of injected electron
- Via electrolyte, to counter electrode

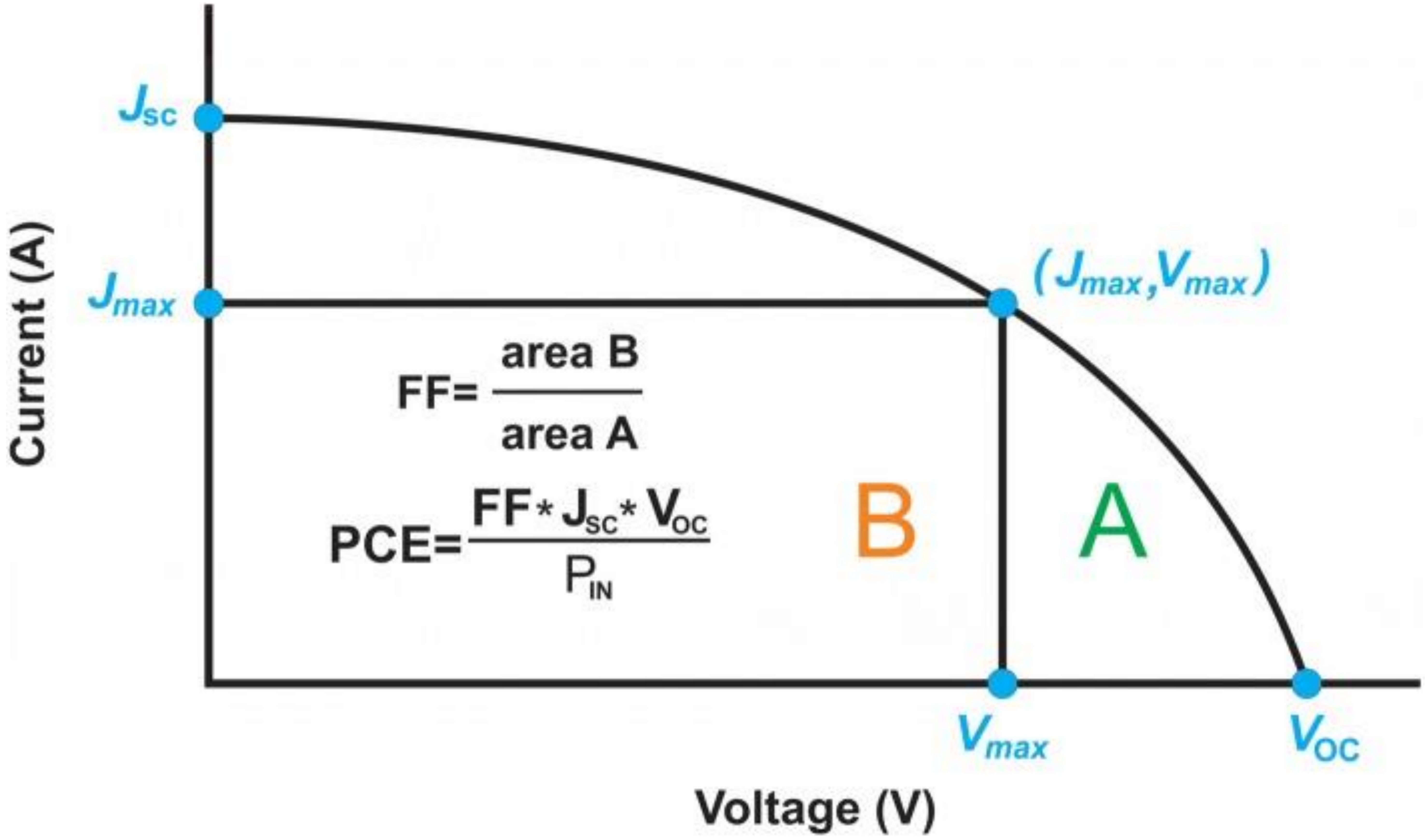
Step 4: Charge Collection

- Reduction at the counter electrode
- Dye regeneration to ground state

Performance Characteristics

- Short Circuit Current (J_{sc})
- Open Circuit Voltage (V_{oc})
- Incident Photo to Current Conversion Efficiency (IPCE)
- Maximum Power Output (P_{max})
- Fill Factor (FF)
- Overall Efficiency (η)





$$IPCE = 1240 \times J_{sc} / P_{in} \cdot \lambda$$

$$FF = J_{max} \times V_{max} / J_{sc} \times V_{oc}$$

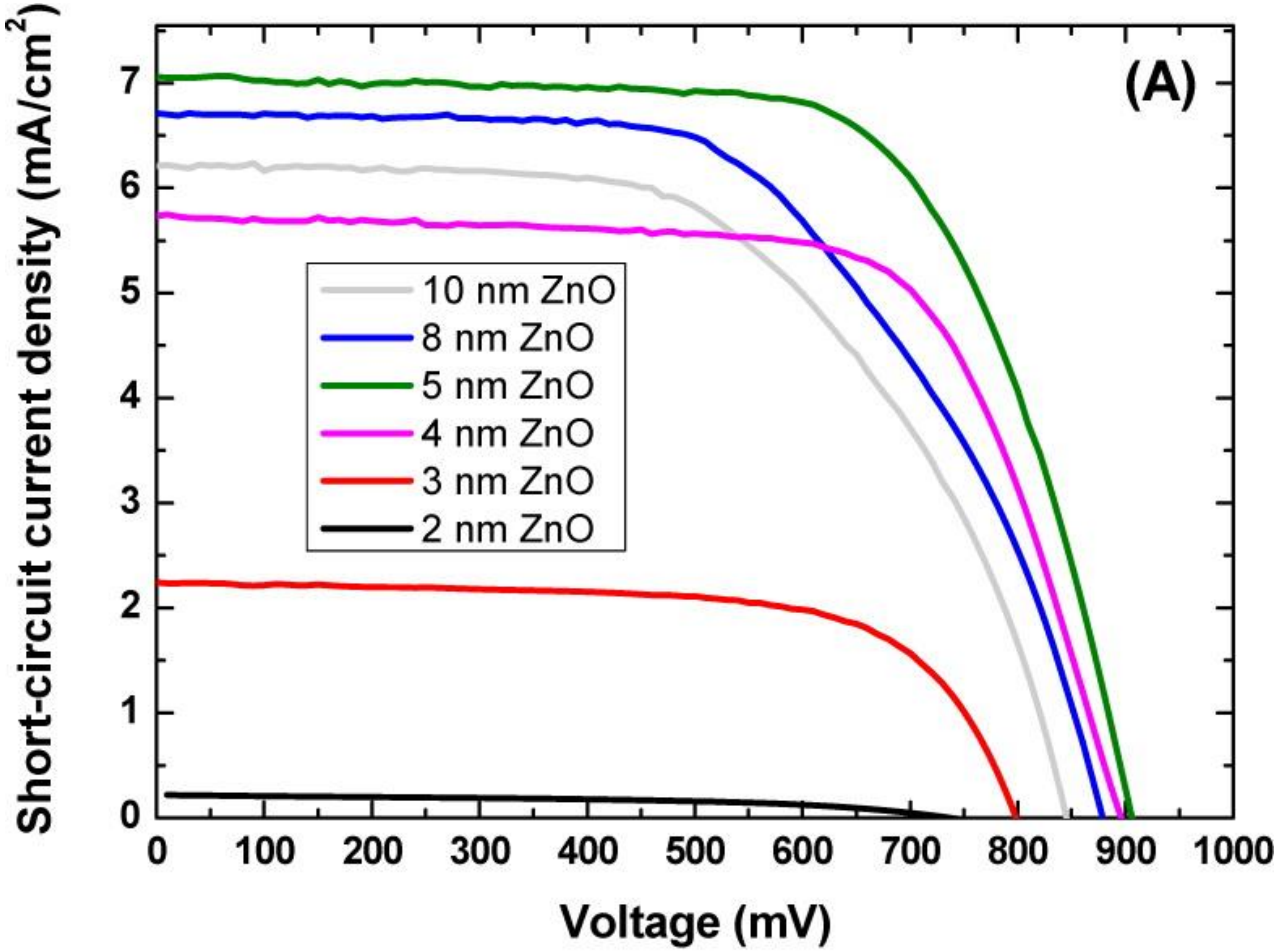
$$P_{max} = J_{max} \times V_{max}$$

$$\eta = J_{sc} \times V_{oc} \times FF / P_{in}$$

Prihanto Trihutomo *et al.* Journal of Photoenergy 2019

Variance of Performance Characteristics with Thickness of ZnO Film

Thickness of ZnO (nm)	J_{SC} (mA/cm^2)	V_{OC} (mV)	Fill Factor (%)	Efficiency (%)
2	0.22	725.6	49.6	0.08
3	2.3	798.0	67.0	1.2
4	5.9	896.4	68.8	3.6
5	7.2	905.9	67.2	4.4
8	6.8	878.8	58.2	3.5
10	6.3	745.2	57.3	3.1



Vinod Kumar *et al.* Thin Solid Films 2013

Challenges

- Highest reported efficiency (~7.5%) is considerably lower than that of TiO_2-based DSSCs (~12.3%)
- Non uniform sensitisation of the working electrode
- Liquid electrolytes are prone to expansion at elevated temperatures
- Energy mismatch between the oxidised dye and electrolyte
- Stability failure
    - Extrinsic Stability - Poor sealing
    - Intrinsic Stability - A stable temperature of 80^{o} C and a stable operation for 1000 hours

Hypothesis

Incorporation of multi-component semiconductor composites instead of single semiconductor

- TiO_2 based DSSCs have reported the highest efficiency of all DSSCs
- TiO_2 is more stable compared to other metal oxides
- ZnO based DSSCs are cheaper and easier to fabricate

Rationale for Proposed Study

Size-induced Variations in Lattice Dimension, Photoluminescence, and Photocatalytic Activity of ZnO Nanorods
Qiu, X. , Li, L. , Fu, X. , and Li, G - J. Nanosci. Nanotechnol., 8 (2008), 1301–1306

Photocatalytic Activity of ZnO-TiO_2 Hierarchical Nanostructure Prepared by Combined Electro-spinning and Hydrothermal Technique
Kanjwal, M. A. , Barakat, N. A. M. , Sheilkh, F. A., and Kim, H. Y - Macromol. Res., 18 (2010), 233–239

Photocatalyst of TiO_2/ZnO Nano Composite Film: Preparation, Characterization, and Photodegradation Activity of Methyl Orange
Tian, J. T. , Chen, L. J. , Yin, Y. S. , Wang, X. , Dai, J. H. , Zhu, Z. B. , Liu, X. Y. , and Wu, P. W - Surf. Coat. Technol., 204 (2009), 206–214

Sonocatalytic Degradation of Some Dyestuffs and Comparison of Catalytic Activities of Nano-sized TiO_2, nano-sized ZnO and Composite TiO_2/ZnO Powders under Ultrasonic Irradiation
Wang, J. , Jiang, Z. , Zhang, L. Q. , Kang, P. L. , Xie, Y. P. , Lv, Y. H. , Xu, R. , and Zhang, X. D. - Ultrason. Sonochem., 16 (2009), 225–231

Specific Aims

Preparation and Characterisation of ZnO-TiO_2 nano composites

Analysis of Performance Characteristics of Nano composites

Preparation and Characterisation of ZnO-TiO_2 Nano Composites

Deposition of ZnO grains on FTO glass by Spin Coating at 3000 rpm

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Heat Treatment 70 °C for 4 hours

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Hydrothermal Synthesis for deposition of TiO2 at 180 °C for 12 hours

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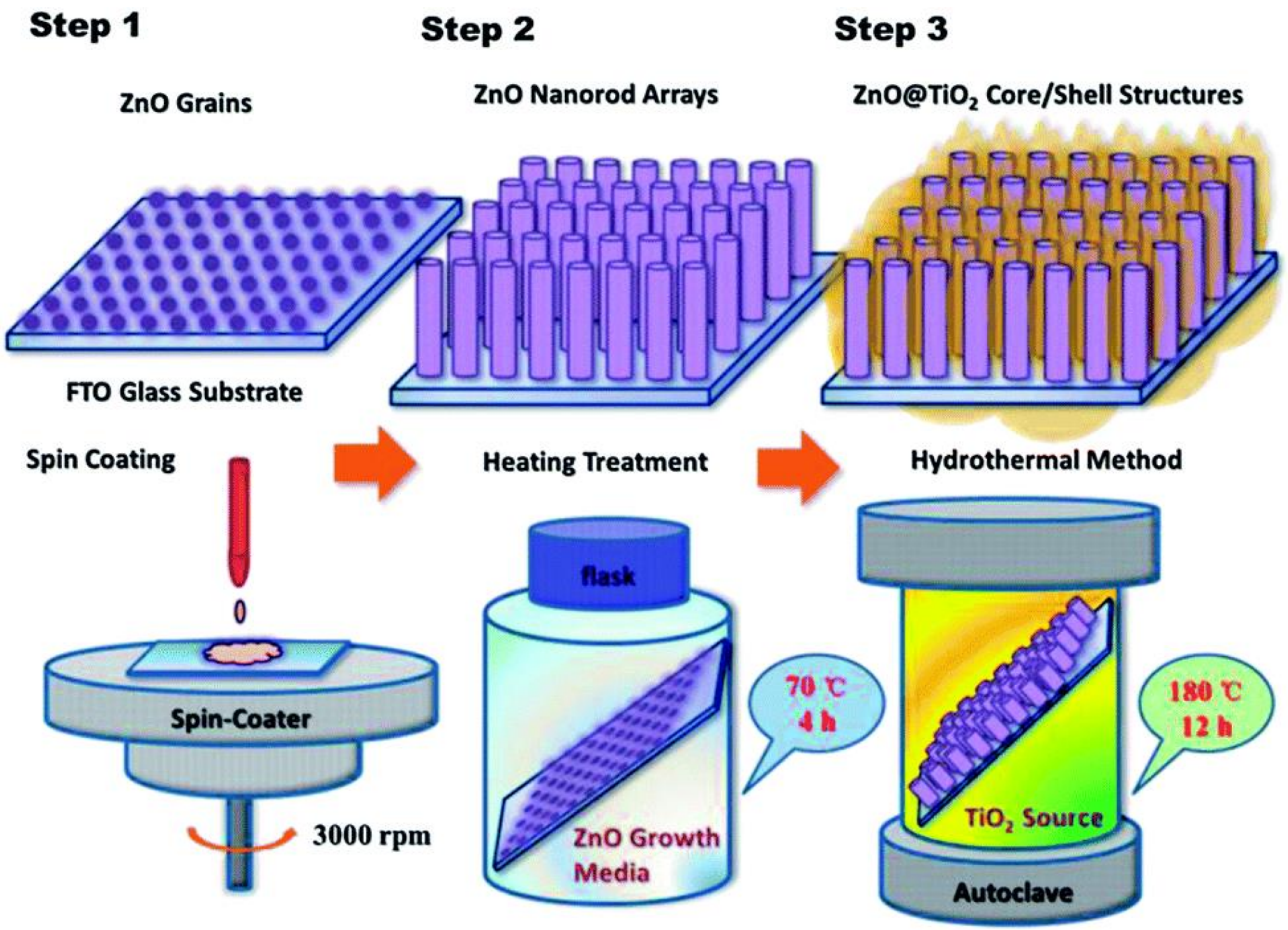
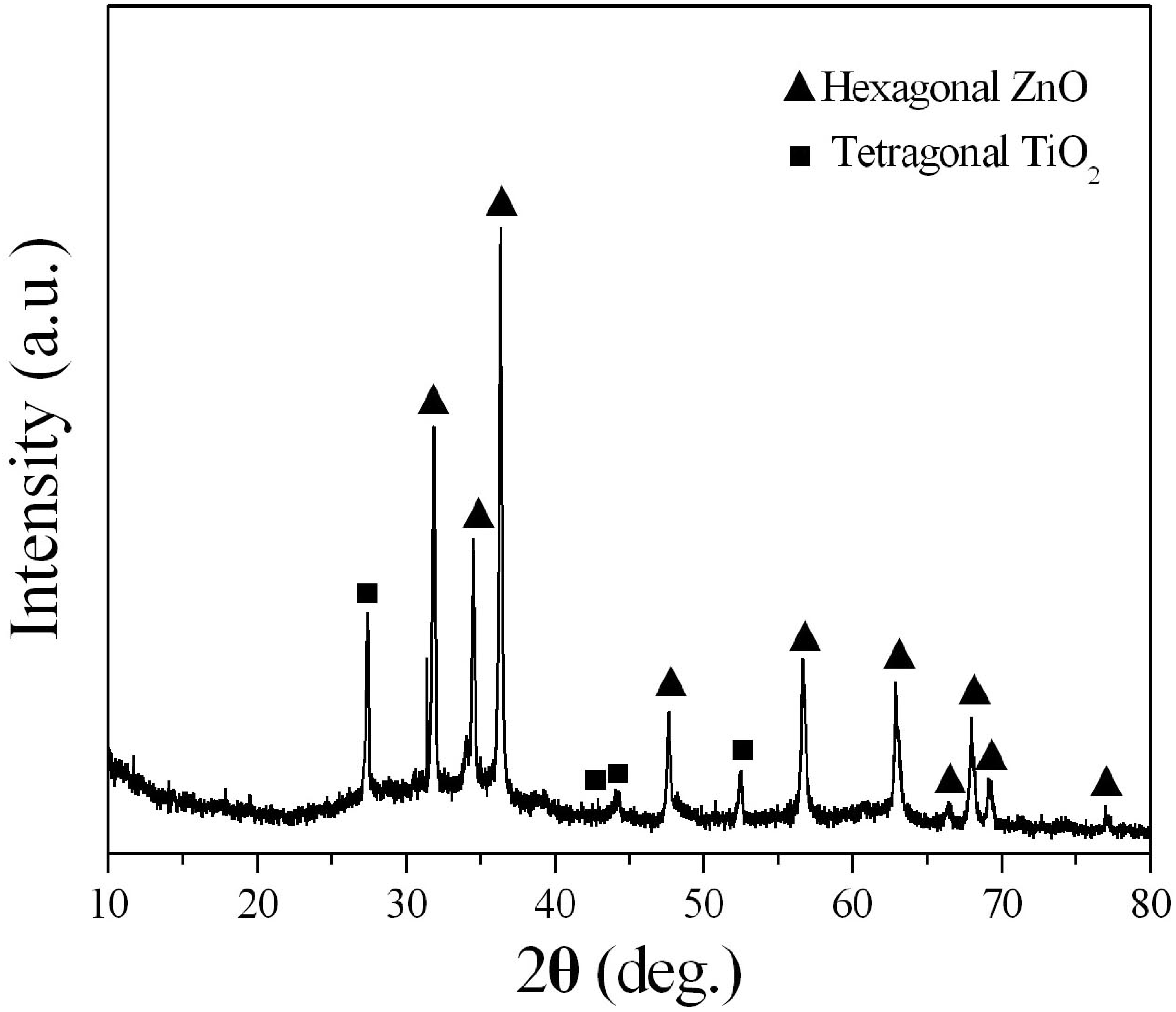
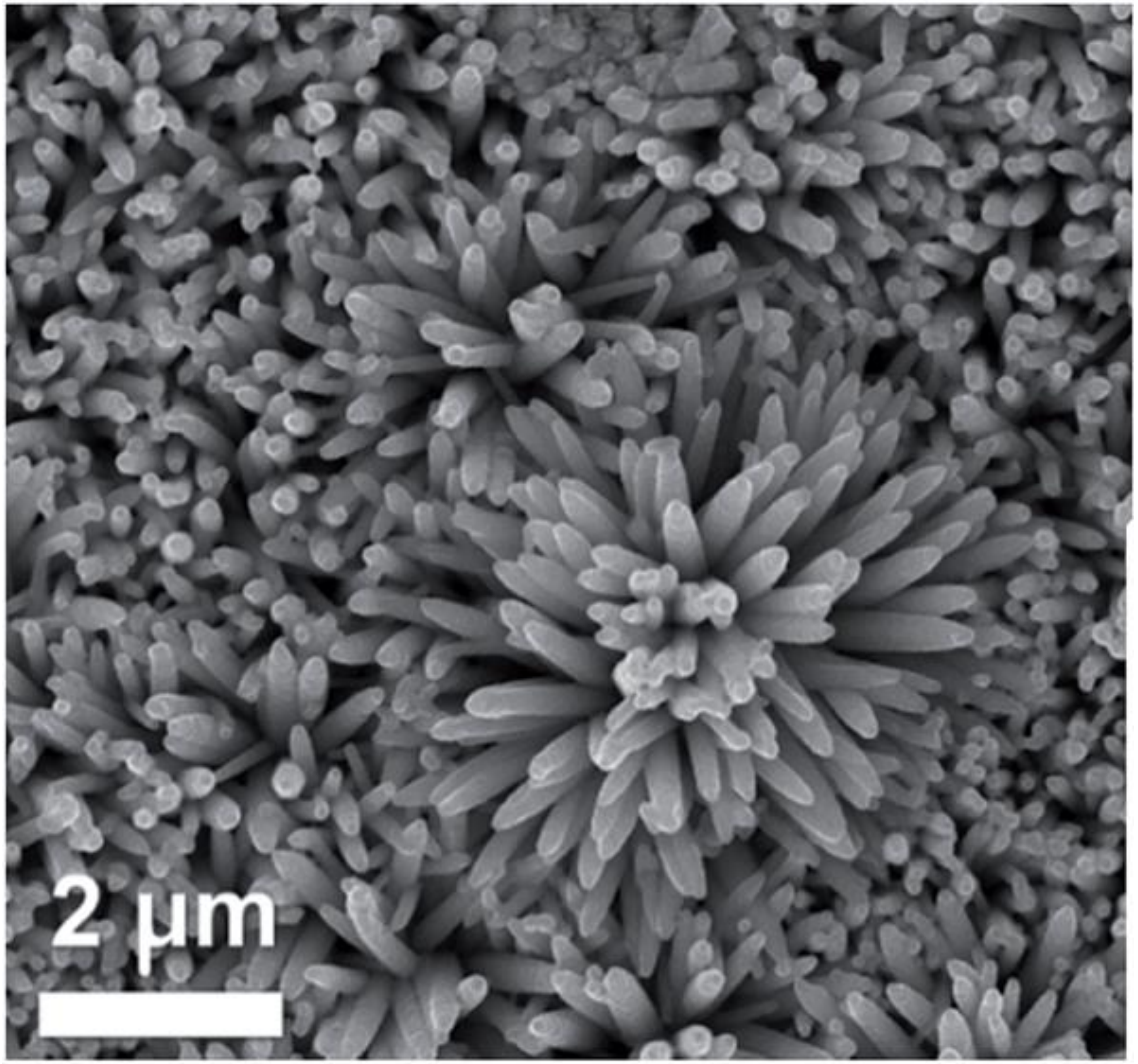
Flushing with deionized water and oven dried





XRD of Zno/TiO_2 Composite Nanorods



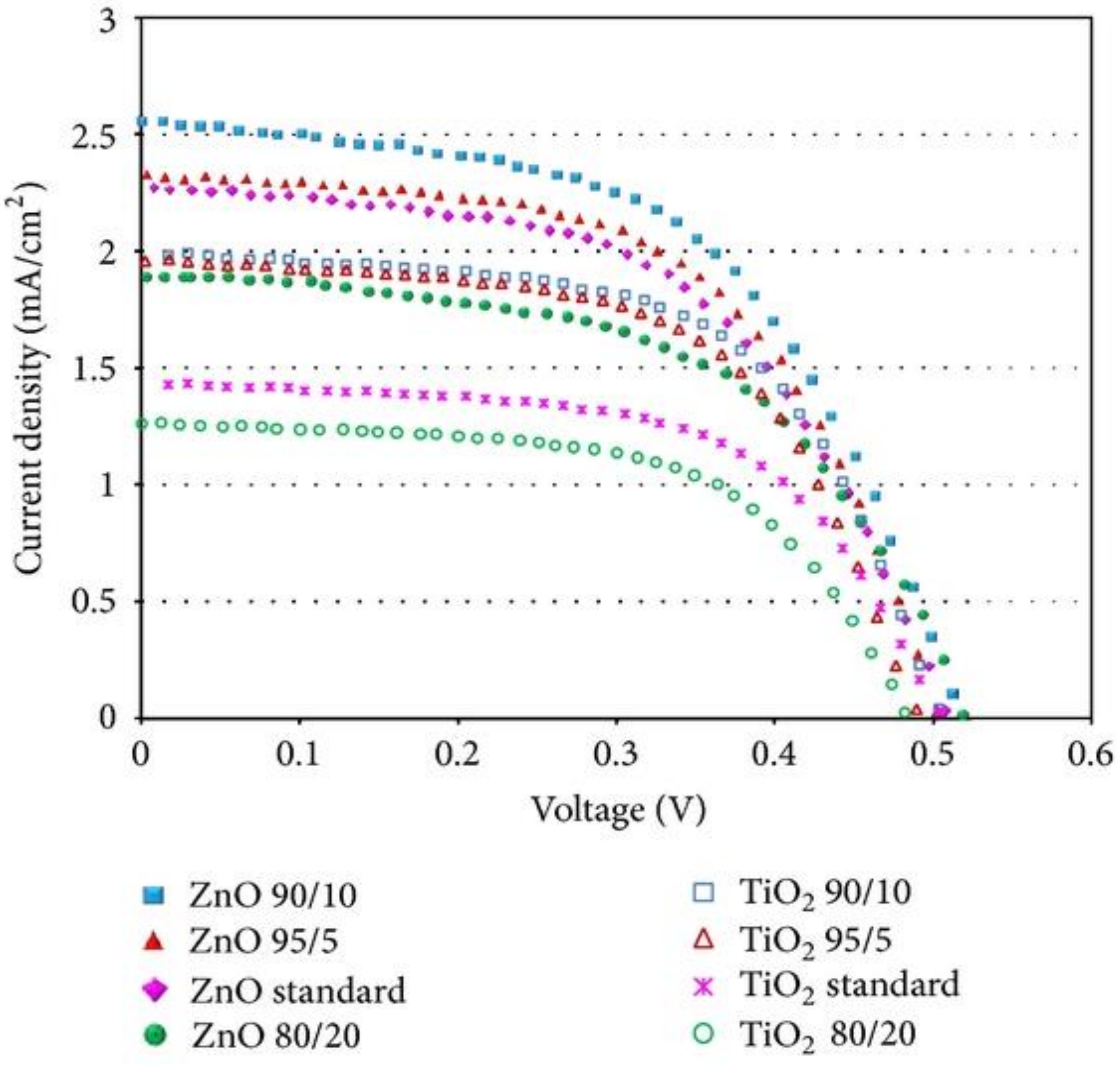
SEM of Zno/TiO_2 Composite Nanorods

L. Z. Pei *et al.* Appl. Phys. 2016

Performance Characteristics of ZnO-TiO_2 Nano Composites

ZnO/ TiO2 Nanocomposites in a 90/10 ratio

- Efficiency similar to ZnO DSSCs
- Higher Stability
- Higher ratio of TiO_2 degrades efficiency and performance characteristics



Chun Cheng *et al.* Nature 2015

Scientific Impact of Study

- TiO_2 based Dye Synthesised Solar Cells have the highest reported efficiency

- ZnO based Dye Synthesised Solar Cells are more cost efficient and easy to fabricate

- Using ZnO/TiO_2 Nanocomposites demonstrates higher stability with no change in efficiency as compared to ZnO based cells

- However, Silicon based Solar Cells continue to have a monopoly in the market due to their long lifespan (>25 years)

Thank You…